What you can do using annotations

OnCommand Insight

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What you can do using annotations

Annotations provide a method for defining information that relates to objects in your environment and then allows you to track objects based on the annotation. For example, you could add building or floor number annotations to devices in your environment and then create a query that returns all of the devices on the first floor of a data center.

Additionally, you might want to look at all devices in a specific data center or business entity and determine which business entity is using the most tier 1 storage. To do this, you assign a data center, business entity, or tier annotation to the device using the OnCommand Insight web UI. Then, you can bring selected user-defined annotations from OnCommand Insight into Data Warehouse. You want to do this to see the annotation values assigned to objects appear in your custom reports.

You can specify which user-defined annotations propagate to Data Warehouse. Annotations are added as additional columns to the object table in the inventory, and to the relevant dimension table in the data marts. When you update the annotations on resources using the OnCommand Insight user interface and initiate or wait for the next Data Warehouse build, you see the results in the following tables:

- dwh_inventory.annotation_value
- dwh_inventory.object_to_annotation

Ensuring annotations entered in OnCommand Insight are included in Data Warehouse requires the following major processes:

- Before you import annotations into Data Warehouse, you must ensure that they are prepared in OnCommand Insight.

  To do this, you can manually run the **Troubleshooting** > **Force Update of Annotations for Data Warehouse** option or wait until the next scheduled transient data run process. When you force the update of annotations, you force the OnCommand Insight server to calculate and place the transient data (such as annotation values) into database tables so that the Data Warehouse ETL process can read the data. The update of annotations data occurs automatically every fifteen minutes; however, you can force it to happen more frequently.

- You then import annotations into Data Warehouse by using the Data Warehouse **Annotations** option.
- If you want to include annotations in reports that you create by using the OnCommand Insight Reporting Portal report authoring tools, you must update the OnCommand Insight reporting metadata model.

When you upgrade Data Warehouse, the annotations job runs automatically during the database restore process. The annotations job runs automatically also when WildFly starts up.

> WildFly is an application server where the OnCommand Insight Java code runs and is needed for both for the OnCommand Insight server and for Data Warehouse.

Preparing annotations in OnCommand Insight

Annotations must be prepared in OnCommand Insight before they can be imported into Data Warehouse.

Steps

1. Log in to the OnCommand Insight Portal as administrator `https://hostname`, where `hostname` is the name of the system where OnCommand Insight is installed.
2. Click on **Admin** > **Troubleshooting**. At the bottom of the page, click on **Advanced Troubleshooting**.
3. In the **Actions** tab, click **Update DWH Annotations (include deleted)**.

Importing user-defined annotations into Data Warehouse

After forcing an annotation update in OnCommand Insight, you need to select the annotations you want in Data Warehouse and initiate a Data Warehouse build. You can wait until the next scheduled build or initiate a build now.

Steps

1. Log in as an administrator to the Data Warehouse Portal at `https://hostname/dwh`, where `hostname` is the name of the system where OnCommand Insight Data Warehouse is installed.
2. From the navigation pane on the left, click **Annotations**.

Annotations

Annotation	Column Name	Target Object	Published
Compute_Resource_Group	Compute_Resource_Group	Virtual Machine	
Data_Center	dataCenter	Host	✔
Data_Center	dataCenter	Storage	✔
Data_Center	dataCenter	Switch	✔
Note	Note	Switch	
Switch_Level	switchLevel	Switch	✔
Tier	Tier	Internal Volume	
Tier	Tier	Qtree	
Tier	Tier	Storage	
Tier	Tier	Storage Pool	
Tier	Tier	Volume	

Edit

The list displays a row for every annotation type and a target object to which the annotation can be assigned. A check mark in the Published column indicates that the annotation was already selected for the particular target object and is already available through the Data Warehouse data marts.

3. Click **Edit** to edit how annotations will be imported from OnCommand Insight.

Edit Annotations

Annotation	Column Name	Target Object	Published All / None	Init With Current All / None
Compute_Resource_Group	Compute_Resource_Group	Virtual Machine	☐	☐
Data_Center	dataCenter	Host	☑	☐
Data_Center	dataCenter	Storage	☑	☐
Data_Center	dataCenter	Switch	☑	☐
Note	Note	Switch	☐	☐
Switch_Level	switchLevel	Switch	☑	☐
Tier	Tier	Internal Volume	☐	☐
Tier	Tier	Qtree	☐	☐
Tier	Tier	Storage	☐	☐
Tier	Tier	Storage Pool	☐	☐
Tier	Tier	Volume	☐	☐

Save Cancel

4. To edit the annotation process, do the following:
   - Select **Published** to add annotations retrieved from OnCommand Insight into the Data Warehouse database. Click **All** to select all annotations on all objects. Click **None** to ensure that all options are not selected.

     > Uncheck this option to remove the annotation column from the specific object's inventory table and associated data marts. If any custom-designed reports use annotation data, the reports do not run successfully.

   - Check **Init with Current** to initialize historical data in Data Warehouse dimension tables with the current annotation value. Click **All** to select all annotations on all objects. Click **None** to ensure that all options are not selected. This check box is disabled after an annotation is published; the check box is enabled for annotations that are not published. For example, if a host is annotated with annotation type "floor" and gets the value "1", and there are 3 rows for that host in the host_dimension table, then selecting **Init with Current** associates the value "1" in the "floor" column for all 3 rows in the host_dimension table. If **Init with Current** is not selected, then only the latest row for that host will have the value "1" in the floor column.
5. Click **Save**.

   A warning message appears indicating that this will cause changes to the structure of the data or data loss, if you are removing annotations.

6. To continue, click **Yes**.

   Data Warehouse initiates an asynchronous annotations job that applies the requested changes. You can see the job in the Jobs page. You can also see the changes in the Data Warehouse database schema.

Viewing the Annotations job in the Jobs list

You can view the Annotations job in the Jobs list and apply the annotation changes to Data Warehouse data marts.

Steps

1. Log in as an administrator to the Data Warehouse Portal at `https://hostname/dwh`, where `hostname` is the name of the system where OnCommand Insight Data Warehouse is installed.
2. From the navigation pane on the left, click **Jobs**.

Displaying annotation changes in the database schema

The database schema reflects the changes in the specific table.

About this task

For example, if you add annotations to a storage array, they appear in the storage or switch table in the inventory or other data marts.

If you update the annotations on resources using the OnCommand Insight user interface and initiate or wait for the next Data Warehouse build, you see a new column added or removed in the corresponding object in inventory (dwh_inventory) and in the corresponding dimension table as well (in the appropriate data mart). You see the results in the following tables:

- dwh_inventory.annotation_value
- dwh_inventory.object_to_annotation

Steps

1. Click on the Data Warehouse tool bar and select **Documentation**.
2. Select **Database Schema**.
3. In the **Database Schema** pane on the left, scroll to the **DWH_INVENTORY** section and click **switch**.

Database Schema

Databases

storage_port
storage_to_applica
switch
switch_port
switch_port_to_app
switch_to_applicati
tape
tape_controller
tape_port
tier
violation
virtual_switch
virtual_to_backend
vm_to_application
volume
volume_in_storage

dwh_inventory.switch

Column	Type	Nullable	Description
id	int(11)	false	GUID of the switch.
fabricId	int(11)	true	GUID of the fabric on which this switch is configured to operate. References: • **id** in dwh_inventory.fabric
identifier	varchar (255)	false	Identifier of the device.
wwn	varchar (255)	false	WWN of the switch.
ip	varchar (255)	false	IP address of the switch.
Name	varchar (255)	false	Name of the switch.
Manufacturer	varchar (255)	true	Manufacturer of the switch
Model	varchar (255)	true	Manufacturer's model of the switch.
Firmware	varchar (255)	true	Firmware version running on the switch.

4. The **dwh_inventory.switch** table reflects the changes:

Database Schema

Databases

host_group_dimen
internal_volume_c
internal_volume_di
qtree_capacity_fac
qtree_dimension
service_level_dim
storage_dimension
storage_pool_dime
tier_dimension
vm_capacity_fact
vm_dimension
volume_fact_curre

dwh_capacity.storage_dimension

Column	Type	Nullable	Description
tk	int(11)	false	TK of this storage array row.
name	varchar (255)	false	Name of the storage array.
identifier	varchar (255)	false	Identifier of the device.
ip	varchar (255)	false	IP address of the storage array.
model	varchar (255)	true	Manufacturer's model of the storage array.
manufacturer	varchar (255)	true	Manufacturer of the storage array.
serialNumber	varchar (255)	true	Serial number for the storage array.
microcodeVersion	varchar (255)	true	Version of the firmware running on the storage array.
family	varchar (255)	true	Family name of the storage array (e.g. Clariion, Symmetrix etc).
id	int(11)	true	GUID of the storage array in **dwh_inventory.storage**.

The dataCenter annotation column appears in the storage_dimensions table.

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